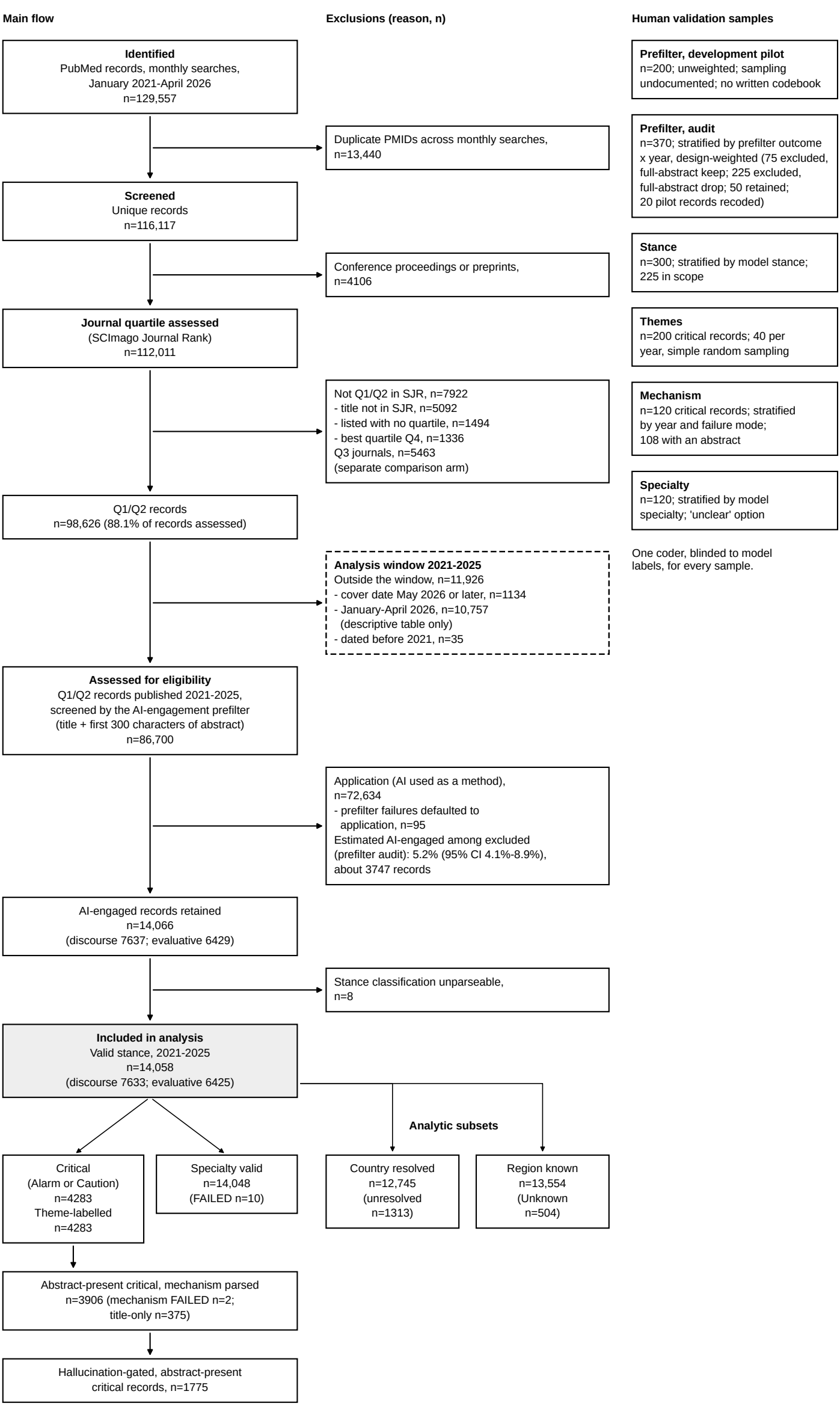
Main flow
Exclusions (reason, n)
Human validation samples
Identified
PubMed records, monthly searches, January 2021-April 2026
n=129,557
Duplicate PMIDs across monthly searches, n=13,440
Screened
Unique records
n=116,117
Conference proceedings or preprints, n=4106
Journal quartile assessed
(SCImago Journal Rank)
n=112,011
Not Q1/Q2 in SJR, n=7922
- title not in SJR, n=5092
- listed with no quartile, n=1494
- best quartile Q4, n=1336
Q3 journals, n=5463
(separate comparison arm)
Q1/Q2 records
n=98,626 (88.1% of records assessed)
Analysis window 2021-2025
Outside the window, n=11,926
- cover date May 2026 or later, n=1134
- January-April 2026, n=10,757
(descriptive table only)
- dated before 2021, n=35
Assessed for eligibility
Q1/Q2 records published 2021-2025, screened by the AI-engagement prefilter (title + first 300 characters of abstract)
n=86,700
Application (AI used as a method), n=72,634
- prefilter failures defaulted to application, n=95
Estimated AI-engaged among excluded (prefilter audit): 5.2% (95% CI 4.1%-8.9%), about 3747 records
AI-engaged records retained
n=14,066
(discourse 7637; evaluative 6429)
Stance classification unparseable, n=8
Included in analysis
Valid stance, 2021-2025
n=14,058
(discourse 7633; evaluative 6425)
Analytic subsets
Critical (Alarm or Caution)
n=4283
Theme-labelled
n=4283
Specialty valid
n=14,048
(FAILED n=10)
Country resolved
n=12,745
(unresolved n=1313)
Region known
n=13,554
(Unknown n=504)
Abstract-present critical, mechanism parsed
n=3906 (mechanism FAILED n=2; title-only n=375)
Hallucination-gated, abstract-present critical records, n=1775
Prefilter, development pilot
n=200; unweighted; sampling undocumented; no written codebook
Prefilter, audit
n=370; stratified by prefilter outcome x year, design-weighted (75 excluded, full-abstract keep; 225 excluded, full-abstract drop; 50 retained; 20 pilot records recoded)
Stance
n=300; stratified by model stance; 225 in scope
Themes
n=200 critical records; 40 per year, simple random sampling
Mechanism
n=120 critical records; stratified by year and failure mode; 108 with an abstract
Specialty
n=120; stratified by model specialty; 'unclear' option
One coder, blinded to model labels, for every sample.